HSLU Lucerne University of Applied Sciences and Arts

Introduction to Substances



CHEM: Chemistry

Dr. Macarena San Martin Ruiz

HS26

Engineering and Architecture

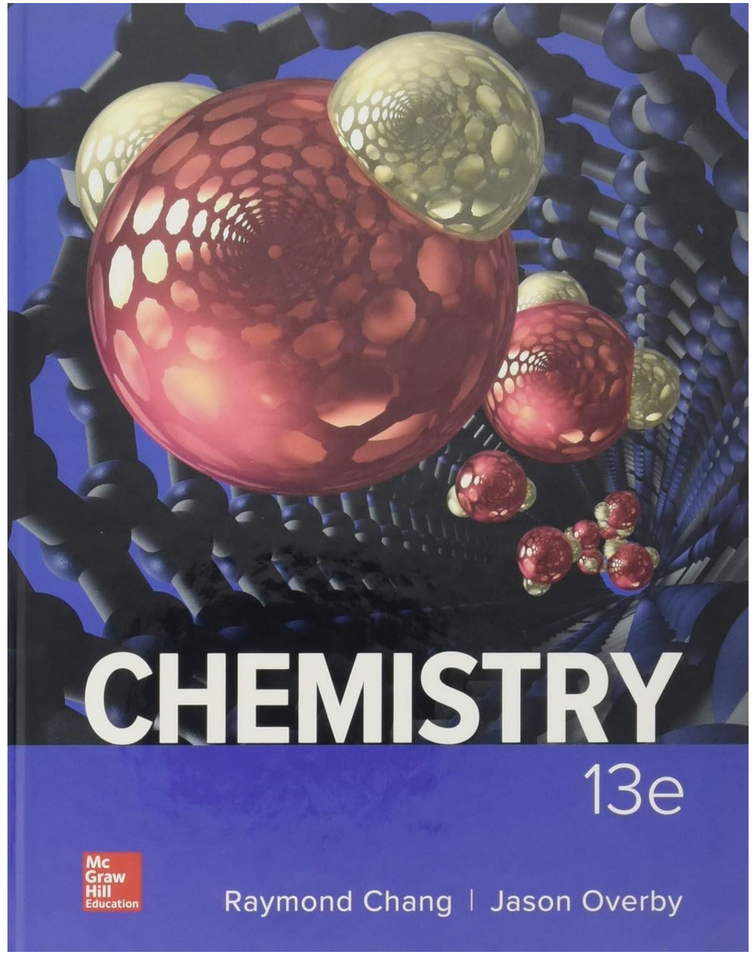
FH Zentralschweiz

Course Details

Lecturer	Date	Week	Topic
Macarena San Martin	16.09.2026	1	Introduction to Matter
Macarena San Martin	23.09.2026	2	Atomic Structure and Radioactivity
Macarena San Martin	30.09.2026	3	Atomic Models (Self Study), Periodic table and electronic configuration
Rebecca Ravotti	07.10.2026	4	Chemical Bonds and Hybridization
Rebecca Ravotti	14.10.2026	5	Lewis Structure
Davide Toniolo	21.10.2026	6	Chemical Reactions
Davide Toniolo	28.10.2026	7	Acids & Bases
Davide Toniolo	04.11.2026	8	Acids & Bases Exercises
Davide Toniolo	11.11.2026	9	Redox Reactions
Davide Toniolo	18.11.2026	10	Redox exercises
Rebecca Ravotti	25.11.2026	11	Metals and Semimetals
Rebecca Ravotti	02.12.2026	12	Organic Compounds nomenclature
Rebecca Ravotti	09.12.2026	13	Organic Exercises
Macarena /Rebecca/Davide	16.12.2026	14	Final Laboratory

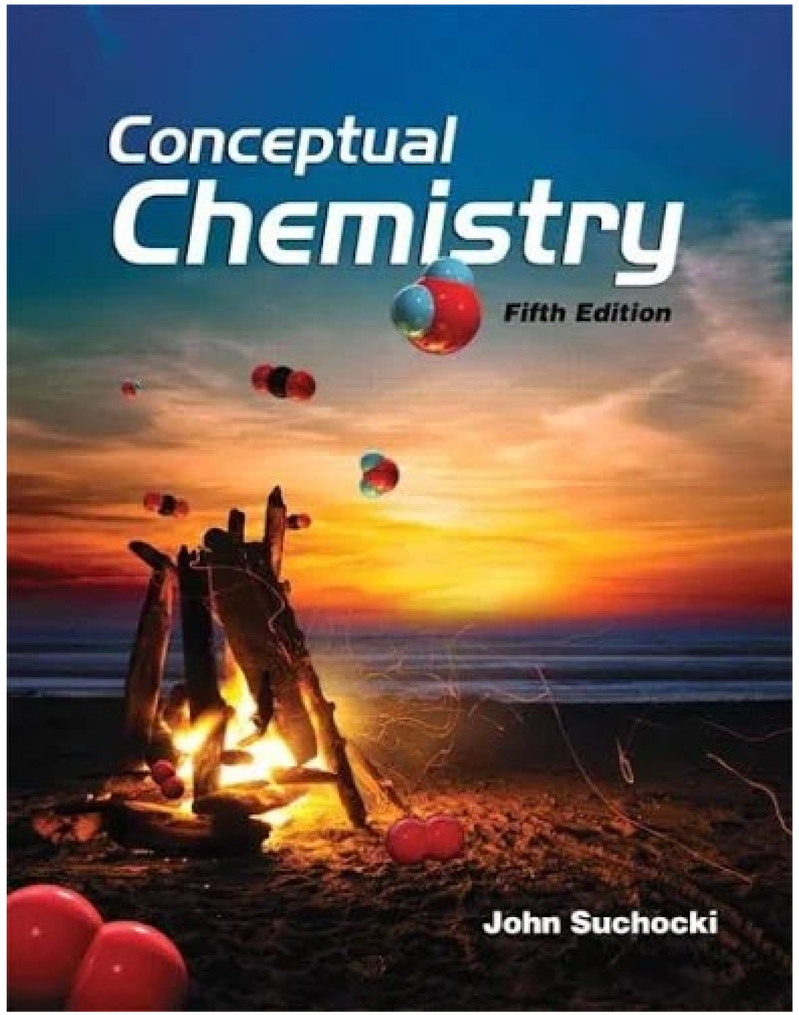
Recommended literature

Chemistry ; Authors, Raymond Chang, Jason Overby ; Edition, **13 or 14th, illustrated** ; Publisher, McGraw-Hill Education, 2018 ; ISBN, 1260085317, 9781260085310



Extra literature

Conceptual Chemistry von John A. Suchocki (ISBN 978-1-292-05558-9)



Details for the exam

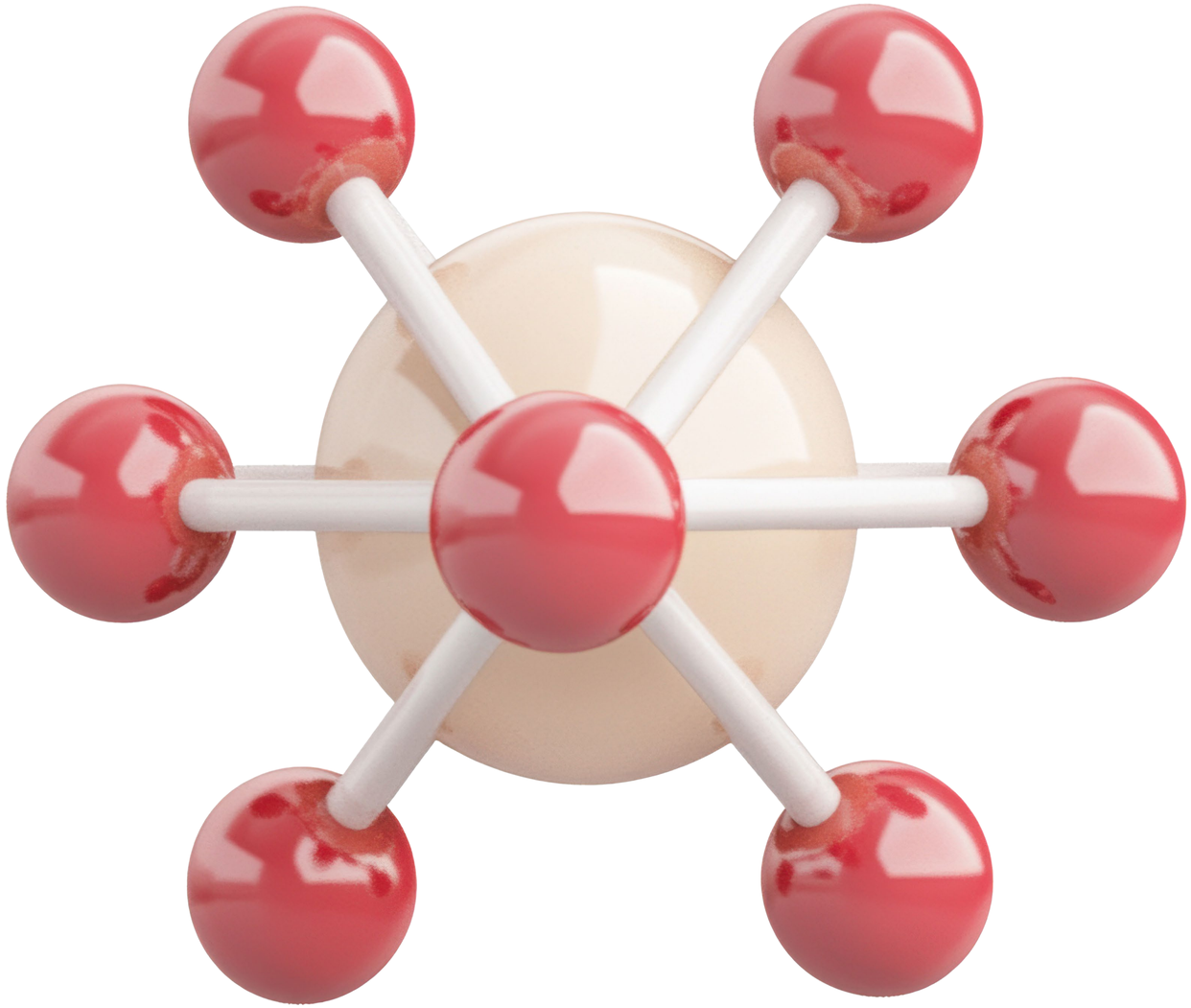
Permitted Aids:

- Only calculators that are not programmable are permitted.
- Writing must be done with a blue or black ballpoint pen. Pencils are not legally valid!
- All other materials will be provided as appendices at the start of the exam, including the periodic table.
- Please check Ilias for the notes that will be available during the exam. They will be uploaded next week.

Structure of the Course

1. Lecture (theory and some exercises)
2. Exercises (guided)
3. Self-study (review slides, repetition using exercises, book, learning videos, etc.)

Breaks: As needed, typically 15 min at 9:40 and 10 min at 10:35- Please tell me if you want an earlier break!





How to Study Chemistry

Compared with other subjects, chemistry is commonly perceived to be more difficult, at least at the introductory level. There is some justification for this perception. For one thing**, chemistry has a very specialized vocabulary**. **At first, studying chemistry is like learning a new language**. Furthermore, some of the concepts are abstract. Nevertheless, with diligence you can complete this course successfully and perhaps even pleasurably.

Listed here are some suggestions to help you form good study habits and master the material:

- Attend classes regularly and take careful notes.
- If possible, always review the topics you learned in class the same day the topics are covered in class. Use the recommended books to supplement your notes.
- Think critically. Ask yourself if you really understand the meaning of a term or the use of an equation. A good way to test your understanding is for you to explain a concept to a classmate or some other person.
- Do not hesitate to ask me for help.

You will find that chemistry is much more than numbers, formulas, and abstract theories. It is a logical discipline packed with interesting ideas and applications.



Today's Content

1) Chemical Substances

- Pure substances and mixtures
- Separation methods
- Physical vs. chemical changes
- Properties of substances (State of matter, density, solubility)

2) Amount of Substance

The mole
Concentration units
Preparing solutions/dilutions

Learning objectives

Know the types of mixtures.

Know some separation methods and be able to explain the general principle

Be able to read important values from the phase diagram

Be able to calculate using the values of relative atomic mass, amount of substance, and concentration (solutions)

Be able to apply the dilution equation

Chapters book Raymond Chang 14th edition:

Chapter 1:

- More in detail 1.7, 1.9

Chapter 3:

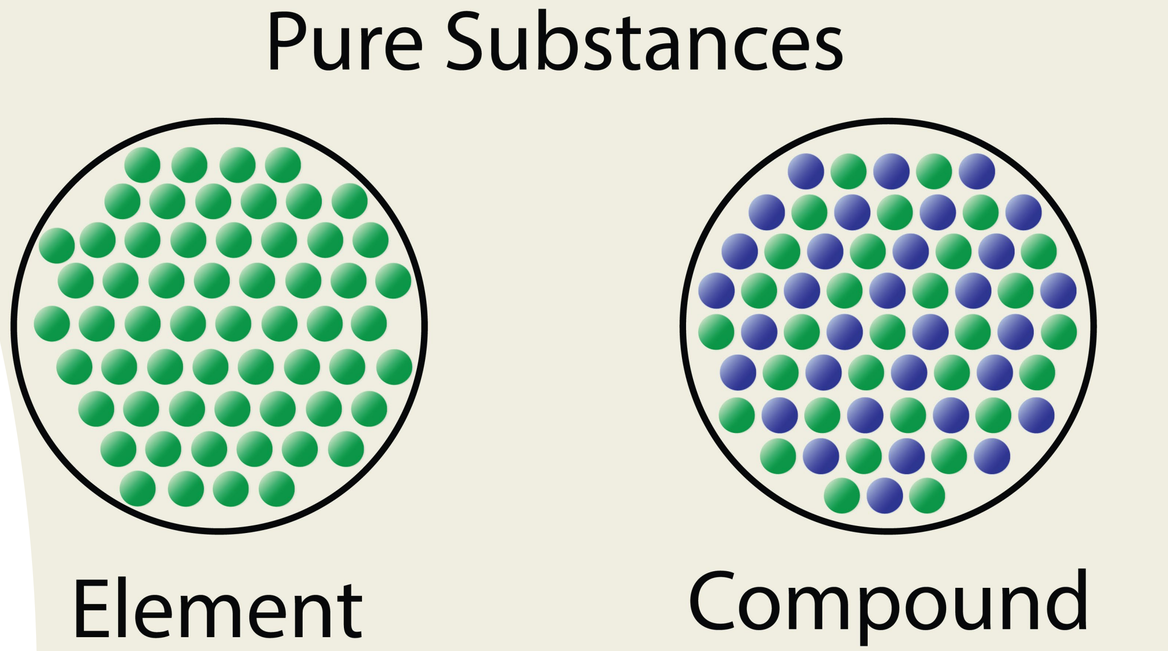
- More in detail 3.1, 3.2, 3.3

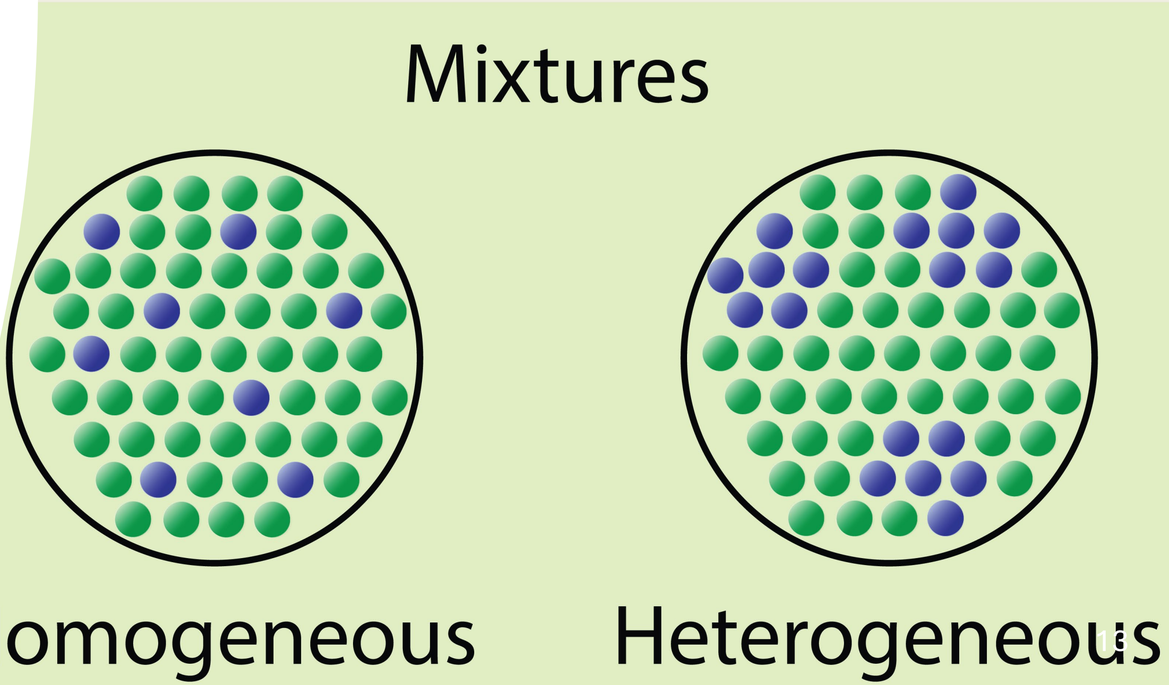
Chapter 5:

- More in detail 5.4

Pure Substances and Mixtures

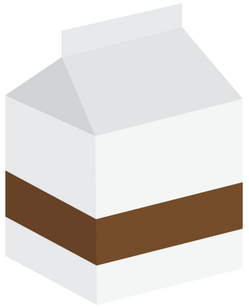
- **Pure substances** have defined properties (e.g., boiling/melting point).
- **Mixtures** vary by ratio and can be homogeneous or heterogeneous.
- Homogeneous: Air (N_2 ~78%, O_2 ~20%)
- Heterogeneous: Milk (visible under microscope).



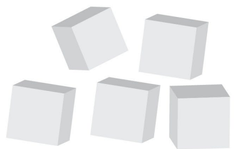




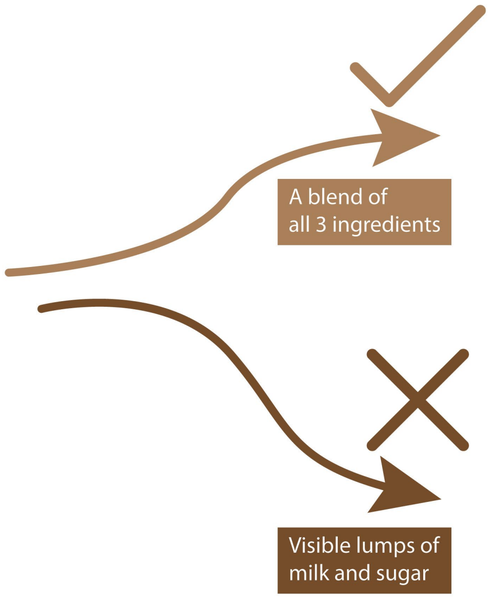
Fresh coffee



Milk



Sugar





Homogeneous



Heterogeneous

Classification of matter

Matter
Occupies space and possesses mass; may exist as solid, liquid or gas

Pure substance
Matter having an invariant chemical composition and distinct properties

Element
Fundamental substance; cannot be separated into simpler substances by chemical methods

Compound
Substance composed of two or more elements in fixed proportions; can be separated into simpler substances and elements only by chemical methods



Gold, Au(s)



Table salt, NaCl(s)

Mixture
Matter consisting of two or more pure substances that retain their individual identities and can be separated by physical methods

Homogeneous
Mixture having a uniform composition and properties throughout

Gas solutions
Gas/gas: air

Liquid solutions
Liquid/gas: soda
Liquid/liquid: vodka
Liquid/solid: sea water

Solid solutions
Alloys
Solid/solid: steel

Heterogeneous
Mixture not uniform in composition and properties throughout

Emulsions
Liquid/liquid: oil and water

Suspensions
Liquid/solid: dirty water

Aerosols
Gas/liquid: fog
Gas/solid: smoke

Solid sols
Solid/solid: granite

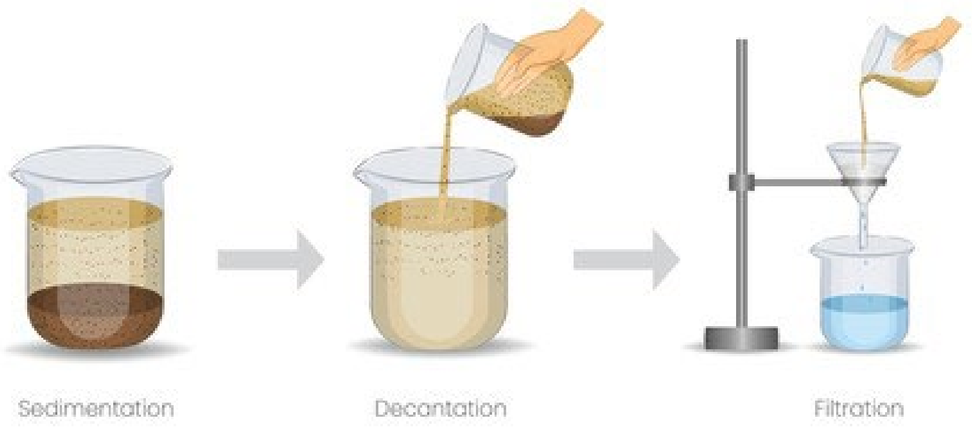
Separation Methods

Based on physical differences like size, density, boiling point.

- Methods: Sedimentation, Filtration, Extraction, Decanting, Distillation.
- Examples:
- - Coffee brewing (extraction)
- - Oil/water (decanting)
- - Alcohol/water (distillation)

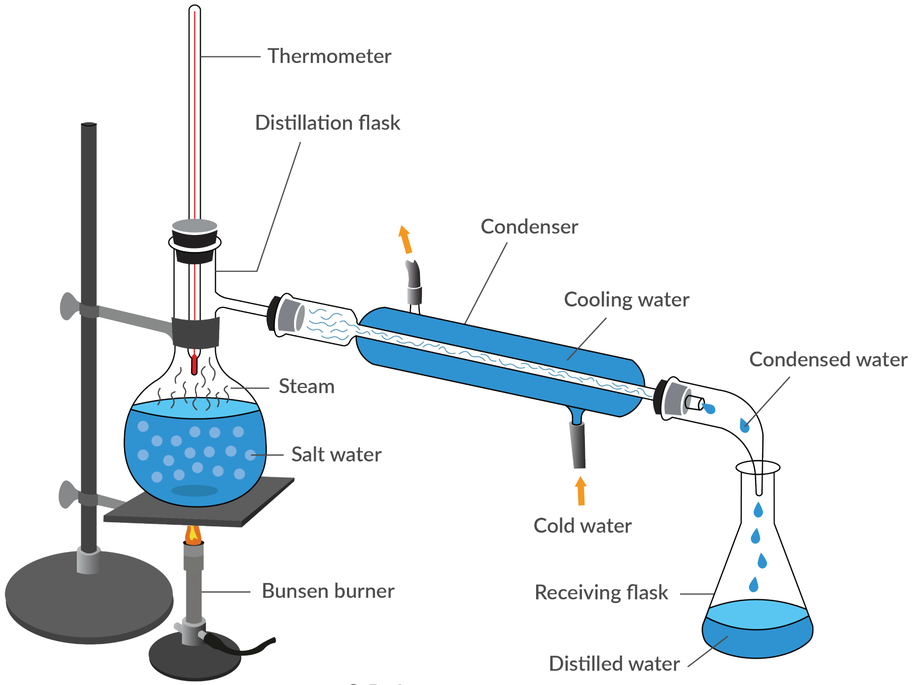






DISTILLATION

Distillation separates mixtures of two or more liquids based on their boiling points. The mixture is heated until one component vaporizes, then cooled in a condenser to collect the **distillate**. The remaining liquid is called the **residue**. This process is also used for purification.



© ReAgent



Sedimentation

Sedimentation is the process by which heavier, insoluble solid particles (suspended solids) settle down at the bottom of a liquid due to gravity.

- Used in water and wastewater treatment

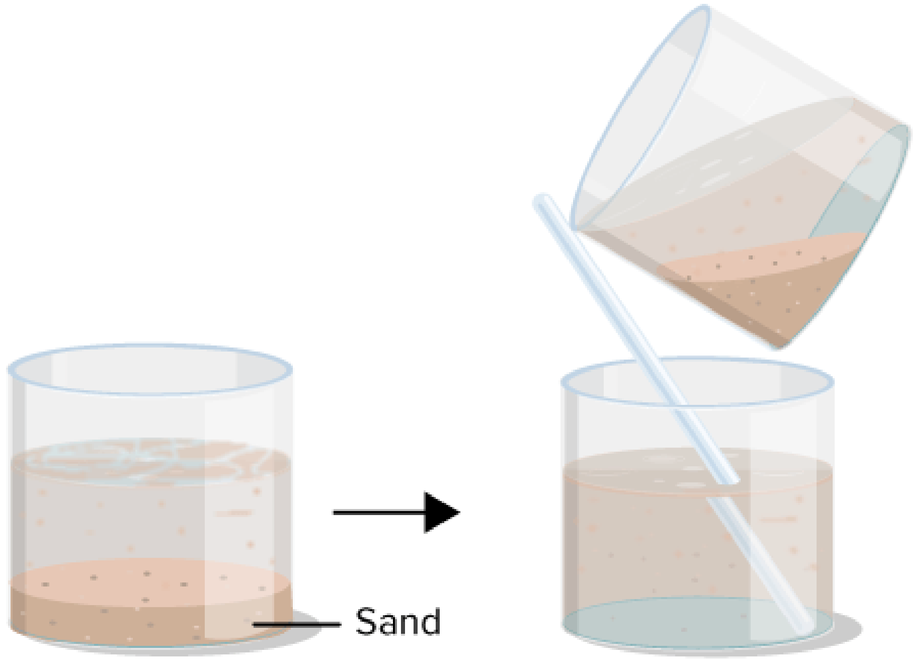


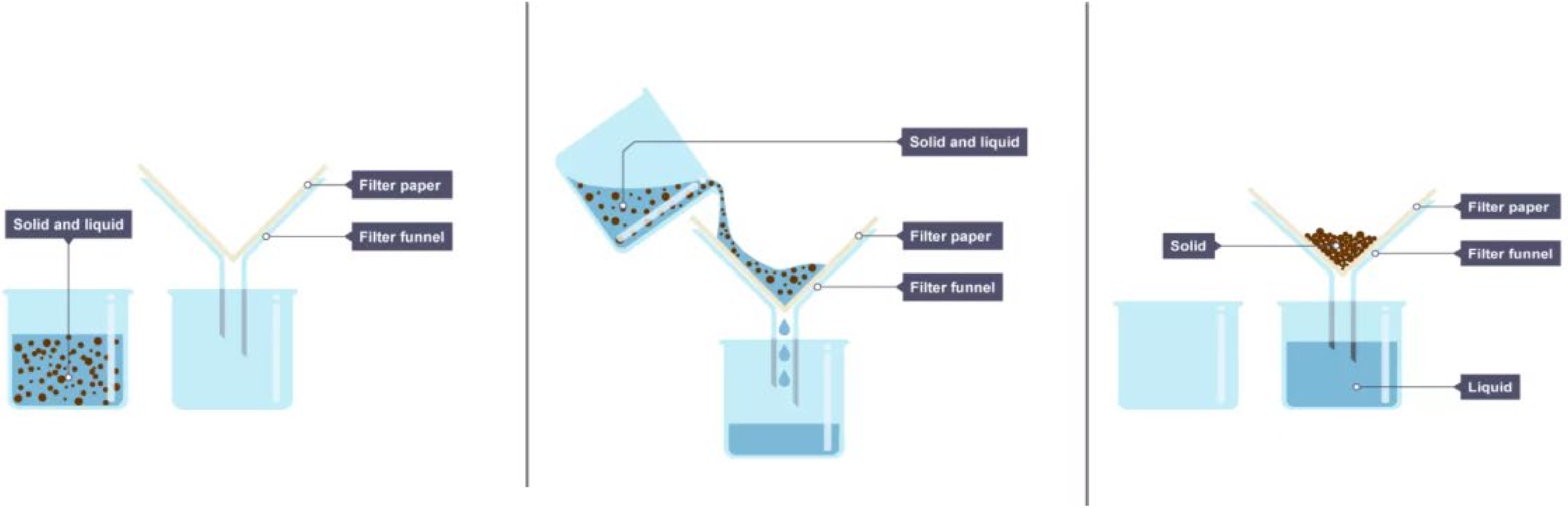
Decantation



Decantation is a process for the separation of mixtures of immiscible liquids or of a liquid and a solid mixture such as a suspension.

Immiscible Liquid Separation: Takes advantage of differences in density of the liquids.A separatory funnel is an alternative apparatus for separating liquid layers. It has a valve at the bottom to allow draining off the bottom layer. It provides better separation between two liquids.Example: Separation of a mixture of oil and water.





FILTRATION

Takes advantage of the physical properties of the state of matter, its size, and solubility in liquid.

Definition:

- Mechanical, physical, or biological operations that separate solids from fluids (liquids or gases) by adding a medium through which only the fluid can pass can be called **filtration**.

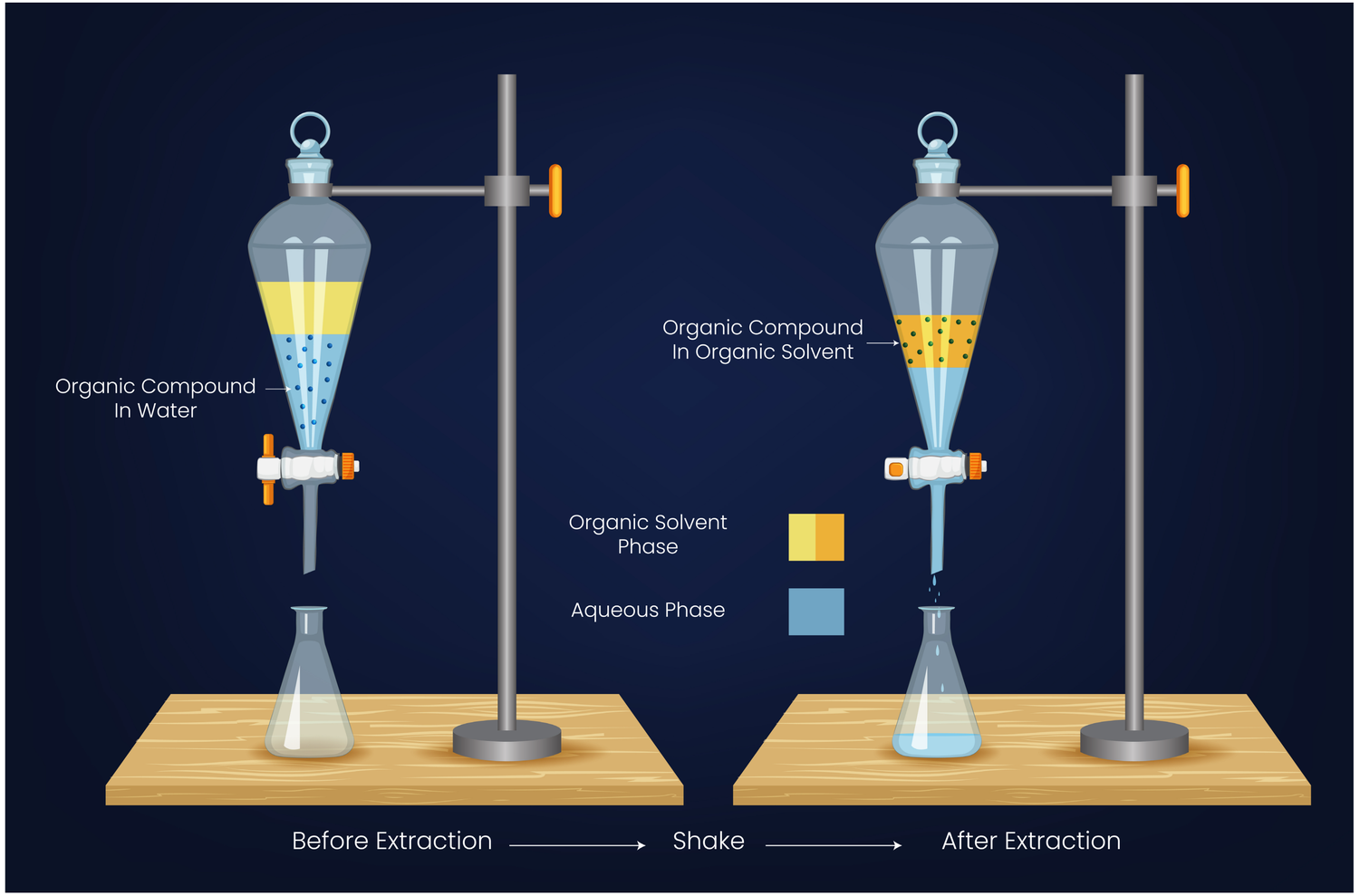
Important

- The substance which is left behind in the filtration medium is called **residue**.
- The liquid which passes through the filtration medium is called **filtrate**.

Extraction

Extraction transfers compounds between phases based on solubility.

For instance, tea's polar flavor compounds dissolve into hot water. Following the "**like dissolves like**" principle, polar compounds favor polar solvents. In a separatory funnel, compounds partition toward the more compatible (soluble) layer.

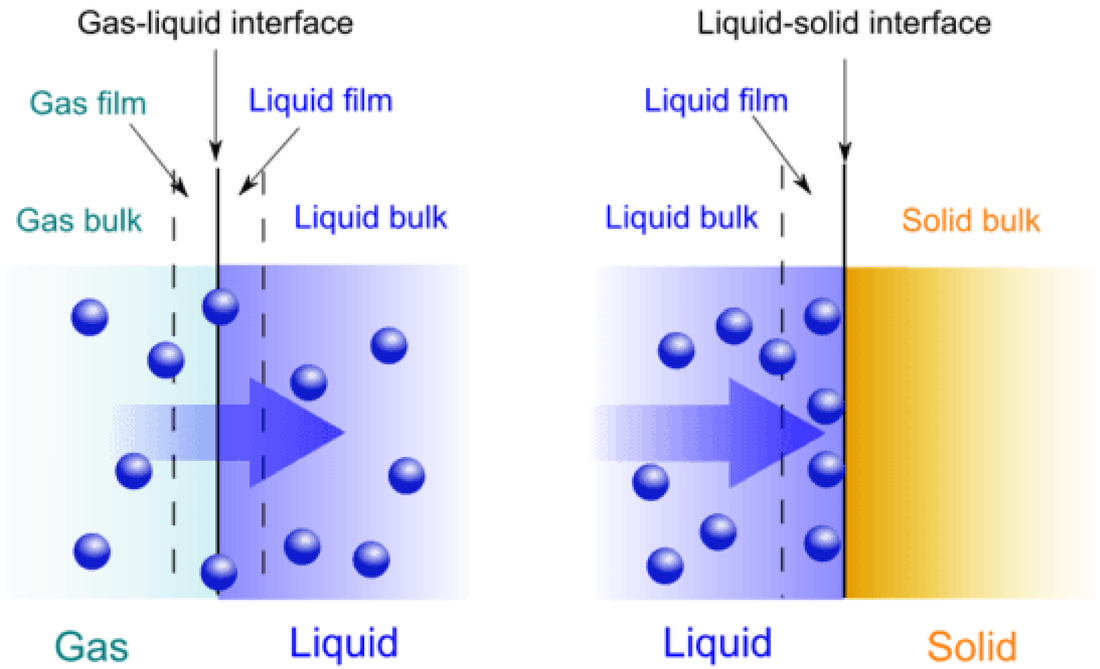


Adsorption

- Activated carbon binds pollutants
- Effective for organics and gases

Absorption

Absorption techniques in environmental engineering involve using a liquid absorbent to selectively dissolve and remove pollutants from a gas stream or wastewater. This process is widely used for CO_2 capture from flue gas, removing odors from wastewater, and addressing other environmental challenges



Absorption Adsorption

DIY: Adsorption – Removing Color from Cola Using Milk

Adsorption:

- **Proteins in the milk** denature due to the **acidity of the cola**.
- **Denatured proteins** form **clumps** and **adsorb** (bind) the **cola coloring** as they **settle down** due to **gravity**.



Exercise: Physical or Chemical Process?



Melting ice



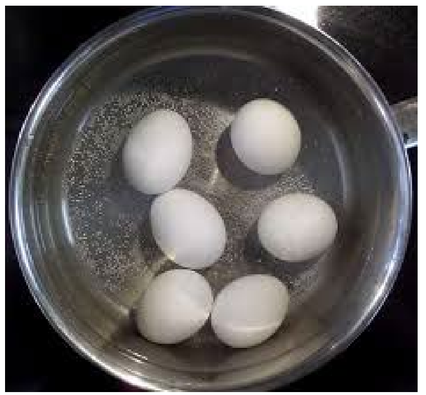
Burning paper



Dissolving salt in water



Brewing coffee



Boiling an egg

Physical vs. Chemical Changes

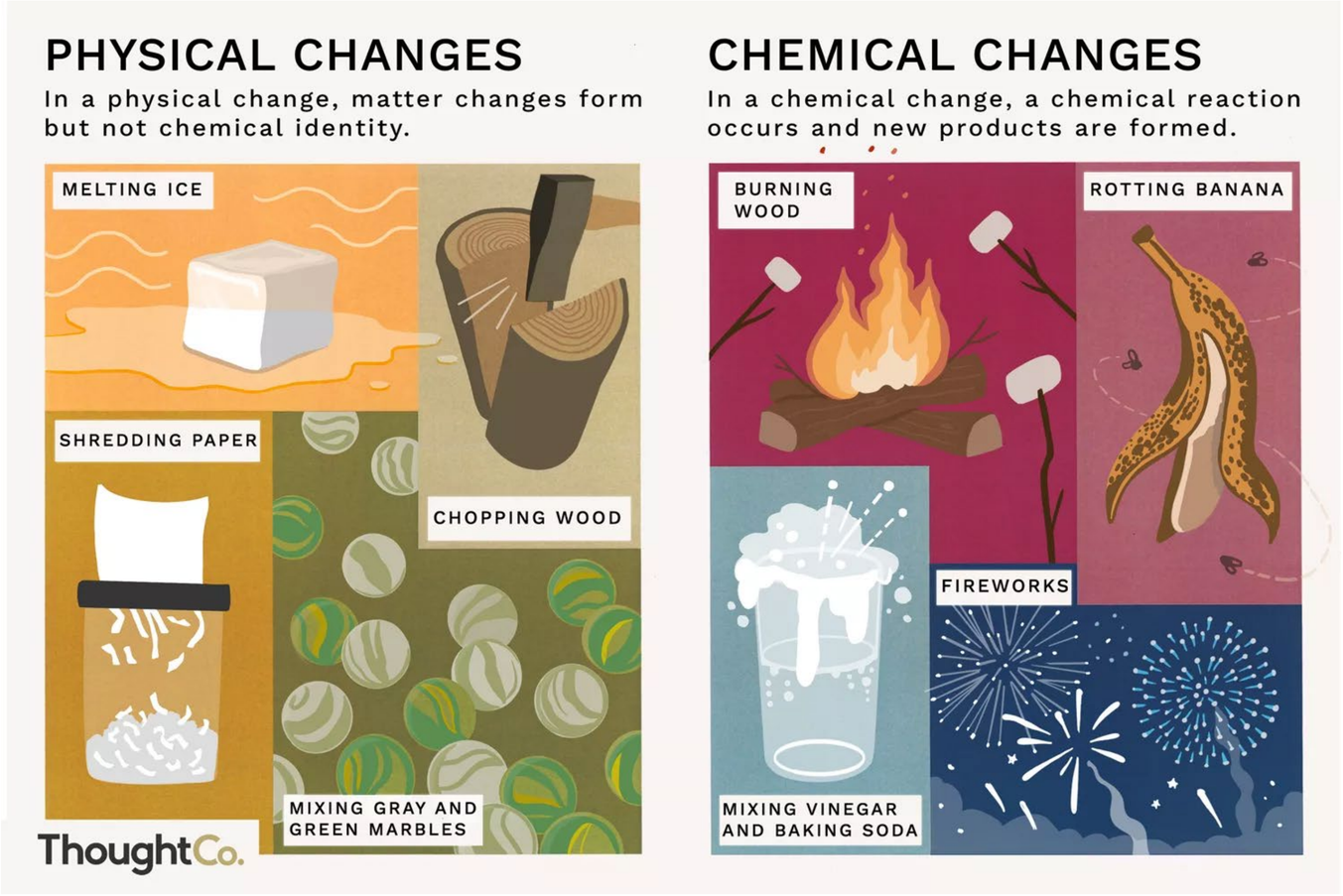
- **Physical:** Changes state/form, no new substance.

Example: Ice melting (still H_2O)

Chemical: New substance formed.

- Example: Propane burning → CO_2 + H_2O + Heat

Physical and Chemical Properties of Matter





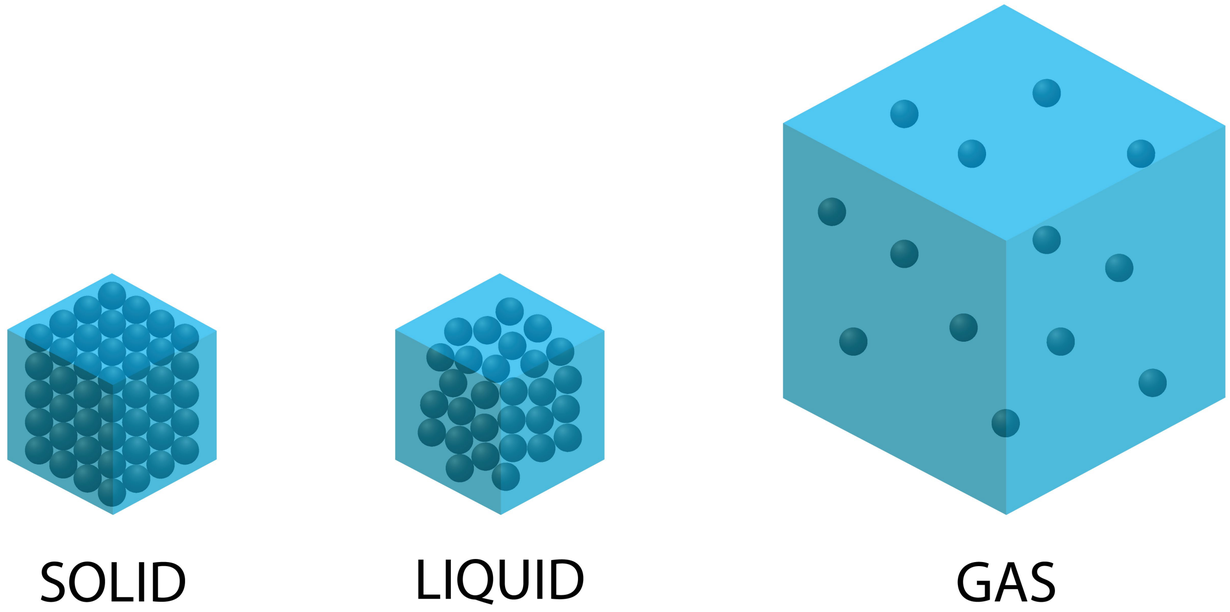


https://www.youtube.com/watch?v=n6wpNhyreDE&t=18s


break
time

States of Matter

- Depends on temperature and pressure.
- Solid, liquid, gas (also plasma).
- Explained with particle model: more order in solids, free movement in gases.
- State changes are reversible and involve energy.

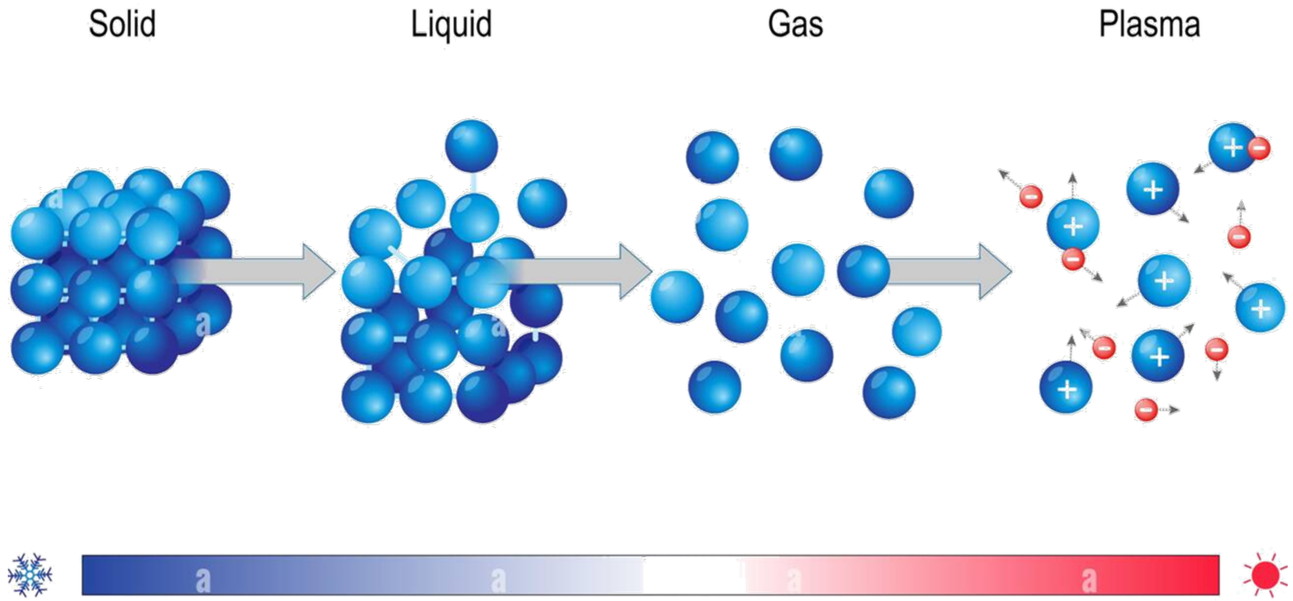


Plasma

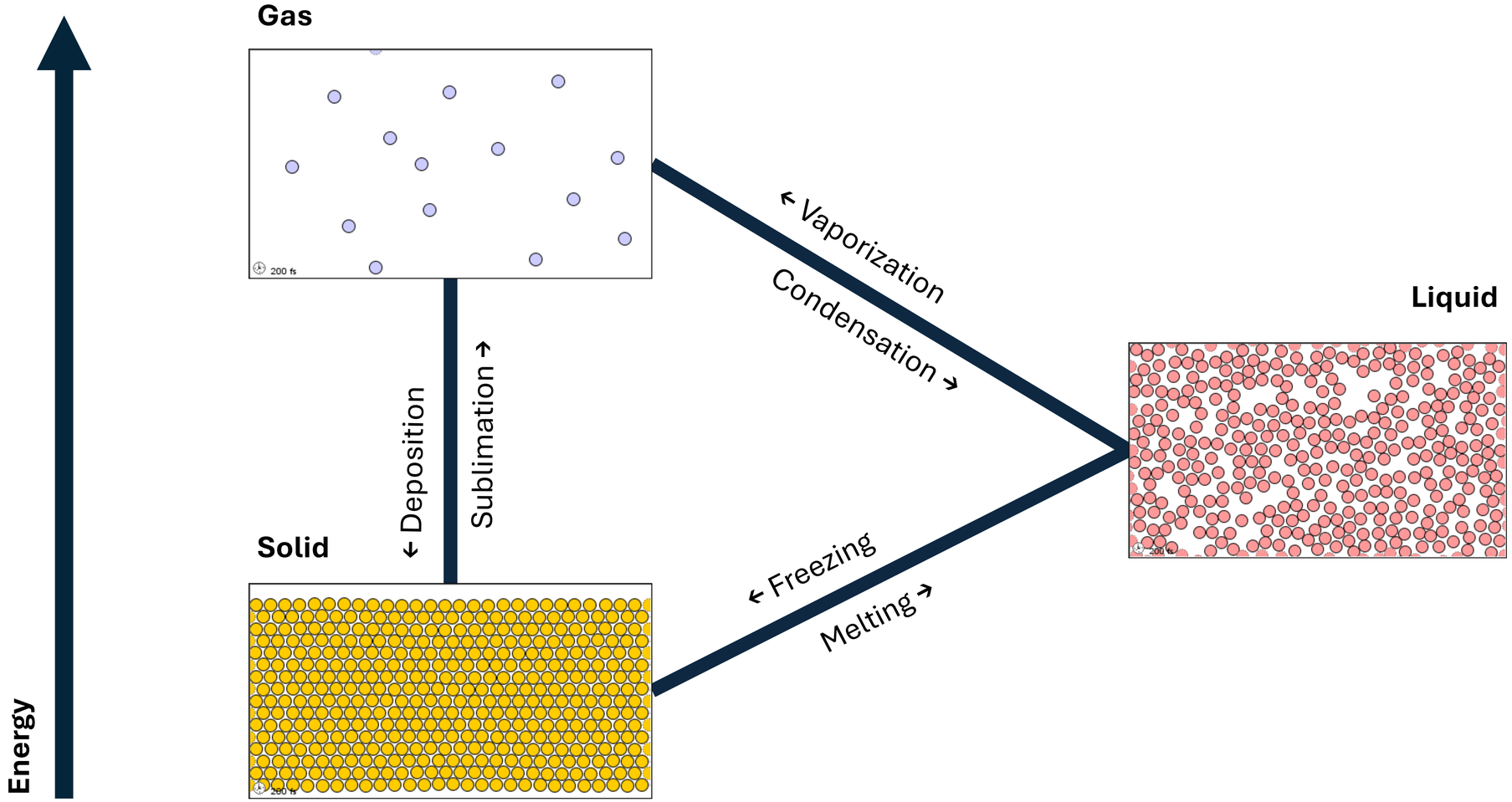
- It contains a significant portion of charged particles (ions and/or electrons)
- Plasma can be artificially generated by heating a neutral gas or subjecting it to a strong electromagnetic field
- The presence of charged particles makes plasma electrically conductive, and it is governed by (and very sensitive to) electromagnetic fields

Examples:

- Stars, including the sun
- Neon signs
- Aurorae
- Lighting



Physical states of matter



Temperature Progression in the Phase Diagram

A phase diagram shows which state of matter (phase) a substance is in, depending on pressure and temperature

Here, using the example of water.

Red Line:1 kg of ice is heated from -40°C to 150°C at no pressure (101 kPa bar).

Blue Line: Ice becomes liquid water by increasing pressu without changing the temperature!

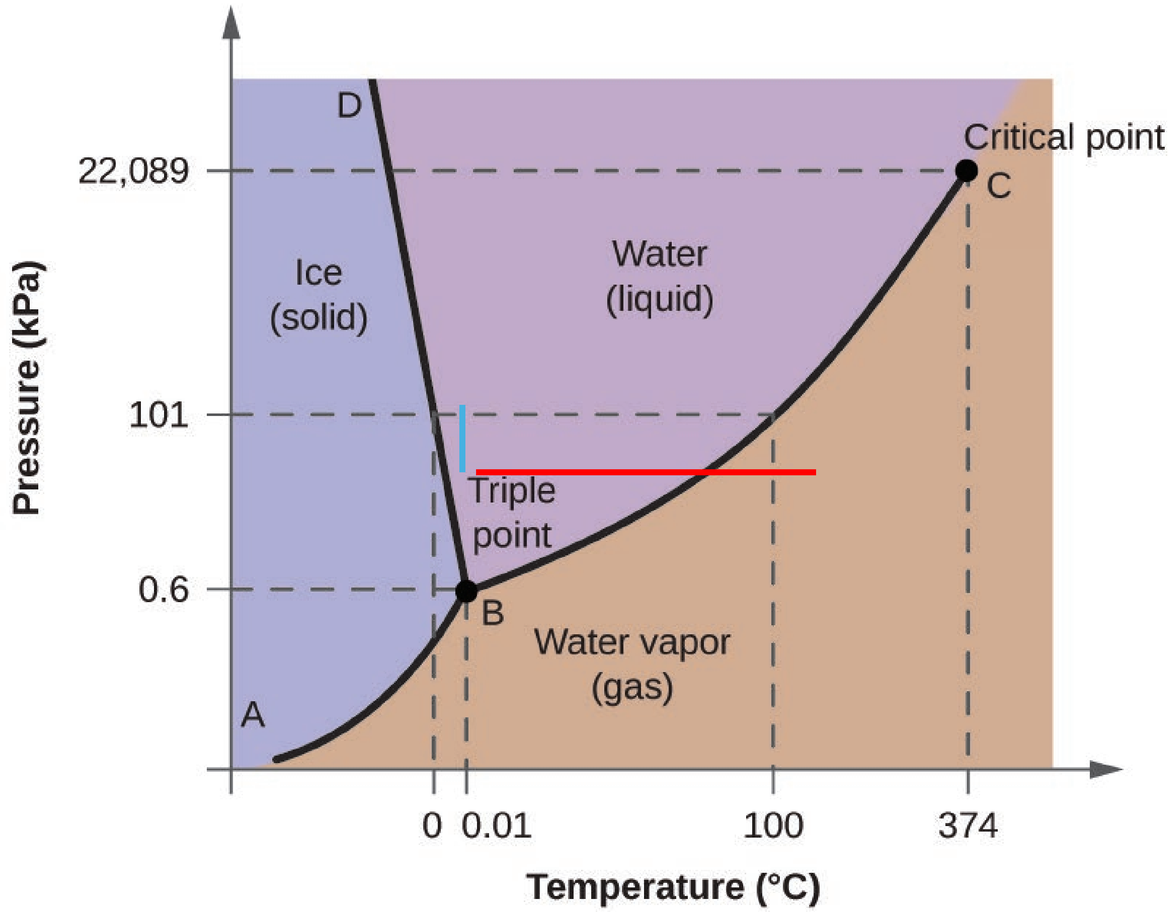
Triple Point: At this specific pressure and temperature, a three phases (solid, liquid, and gas) coexist simultaneou

Phase Diagram Key Regions:

Left area (purple): Ice (solid phase)

Middle area (pink): Liquid water

Right area (brown): Water vapor (gaseous phase)

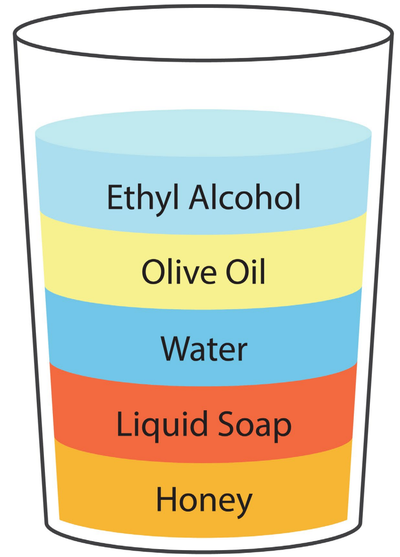


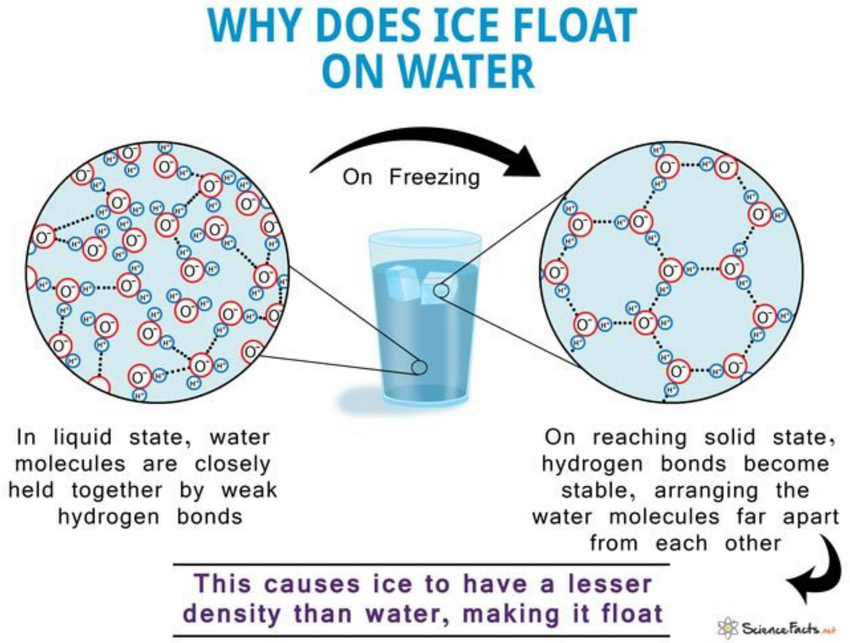
Density

- Density = mass / volume
- Depends on temperature and pressure.
- Water: ~1 g/cm^3, Ice: ~0.92 g/cm^3
- Anomaly: Water has highest density at 4°C, so ice floats

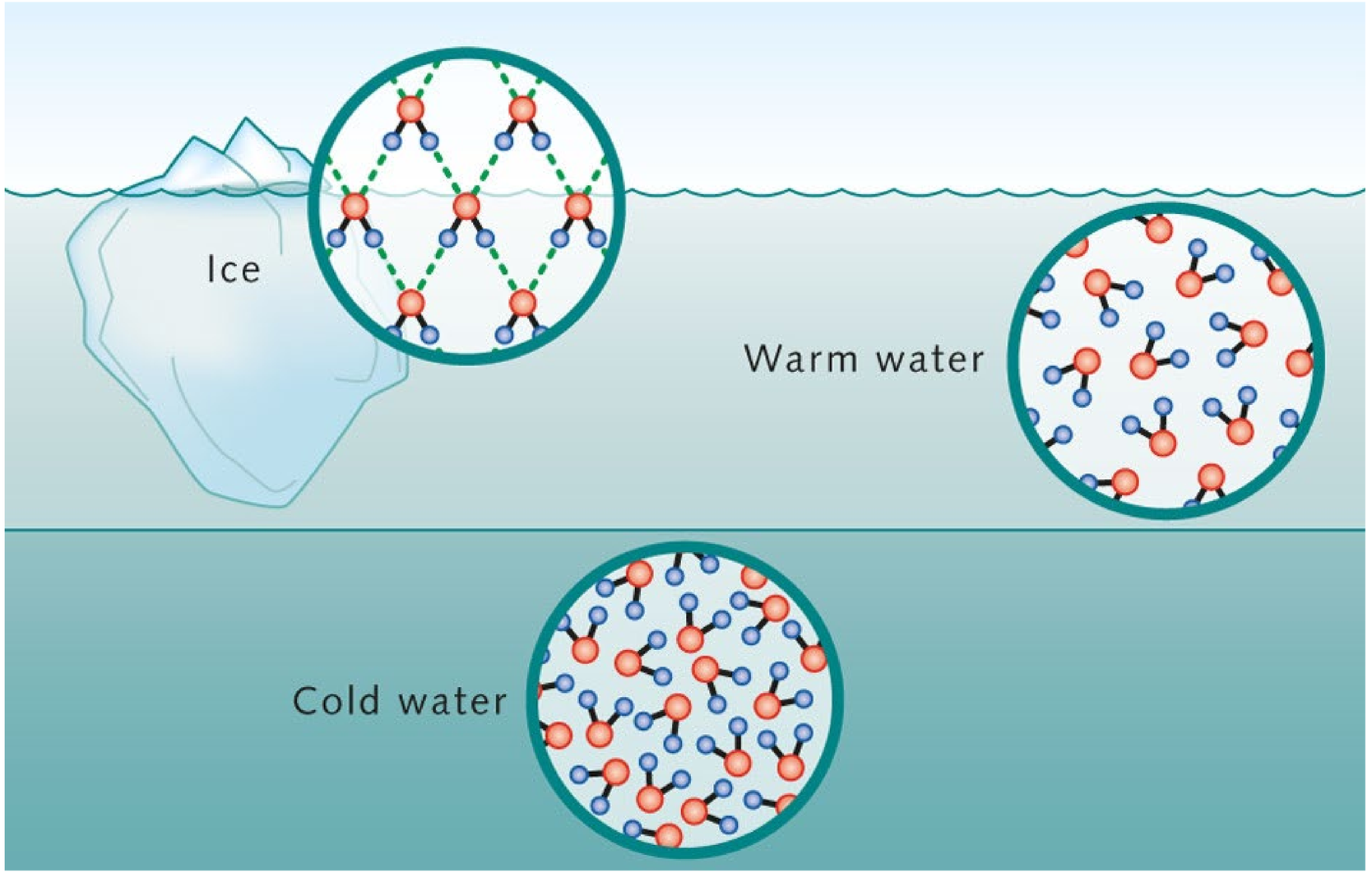
Stoff	Dichte
Gold	19.3 g/cm^3
Wasser	1 g/L
Eis (0°C)	0.91 g/L
Atemluft	0.0012 g/L

13, 19. Se





Density Anomaly of Water



Solubility



Homogeneous mixtures.



Solubility: how much of a substance dissolves before saturation.



Water dissolves salts well (e.g., NaCl).



Involves energy: heat may be released or absorbed.

Solubility

- Solutions = Homogeneous mixtures of substances
- Dissolved means: Every atom/molecule is surrounded by solvent molecules.
- Solubility= The amount of substance that can be dissolved in a solvent.
- Measured in g/L.→ Depends on temperature! Saturated solution= Additional substance precipitates out.

When a substance dissolves, energy is either released or required→ the solution becomes warmer or colder→ Depends on:

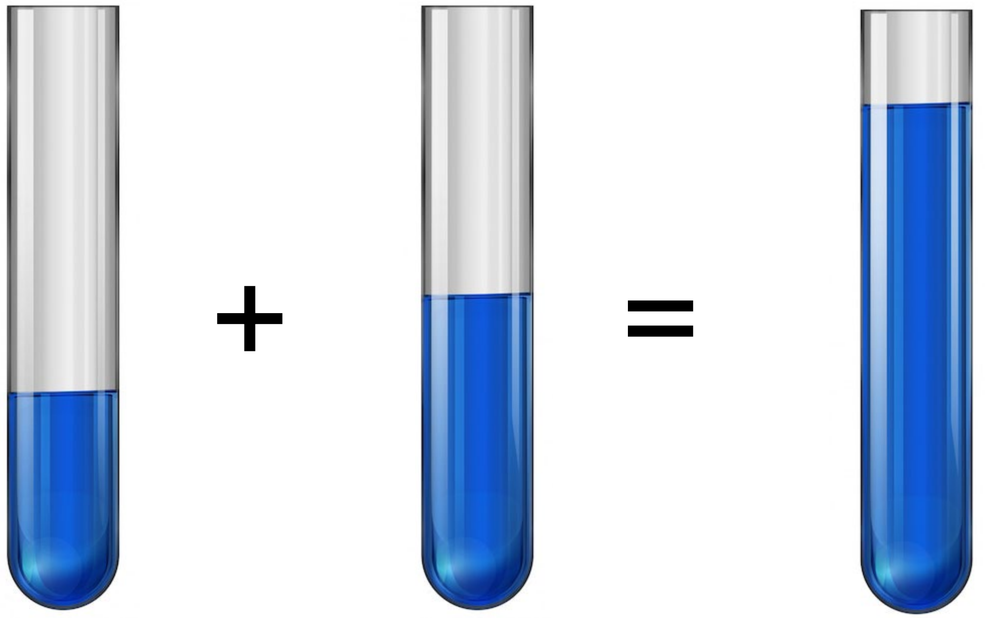
- Lattice energy of the substance
- Bond energy of the solvent
- Solvation energy (in water: hydration enthalpy)
- Exothermic Dissolution Process(Energy is released, solution gets warm):Solvation energy > lattice energy and bond energy
- Isothermic Dissolution Process(Temperature remains constant):Solvation energy = lattice energy and bond energy
- Endothermic Dissolution Process(Energy in the form of heat is required, solution cools down):Solvation energy < lattice energy and bond energy

Gas Solubility

- Affected by pressure (e.g., diving).
- Higher pressure → more gas dissolves in blood.
- Rapid ascent → gas bubbles form (dangerous).
- Helium often replaces nitrogen in diving tanks.

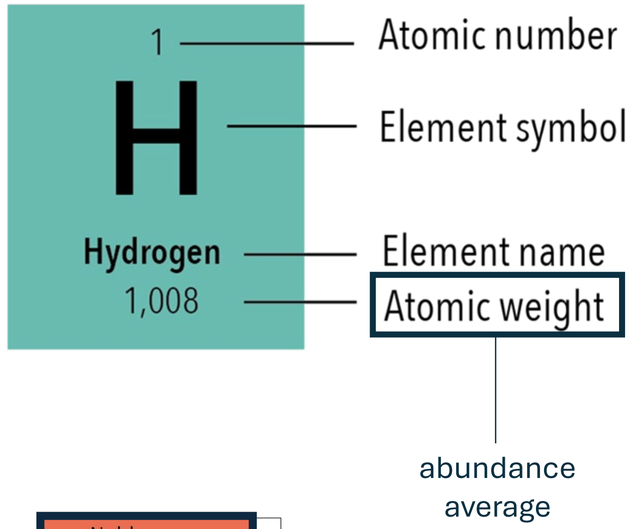


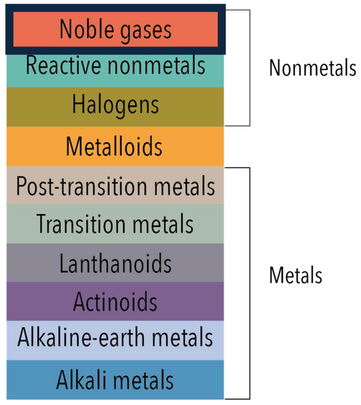
Amount of Substance / Preparing Solutions

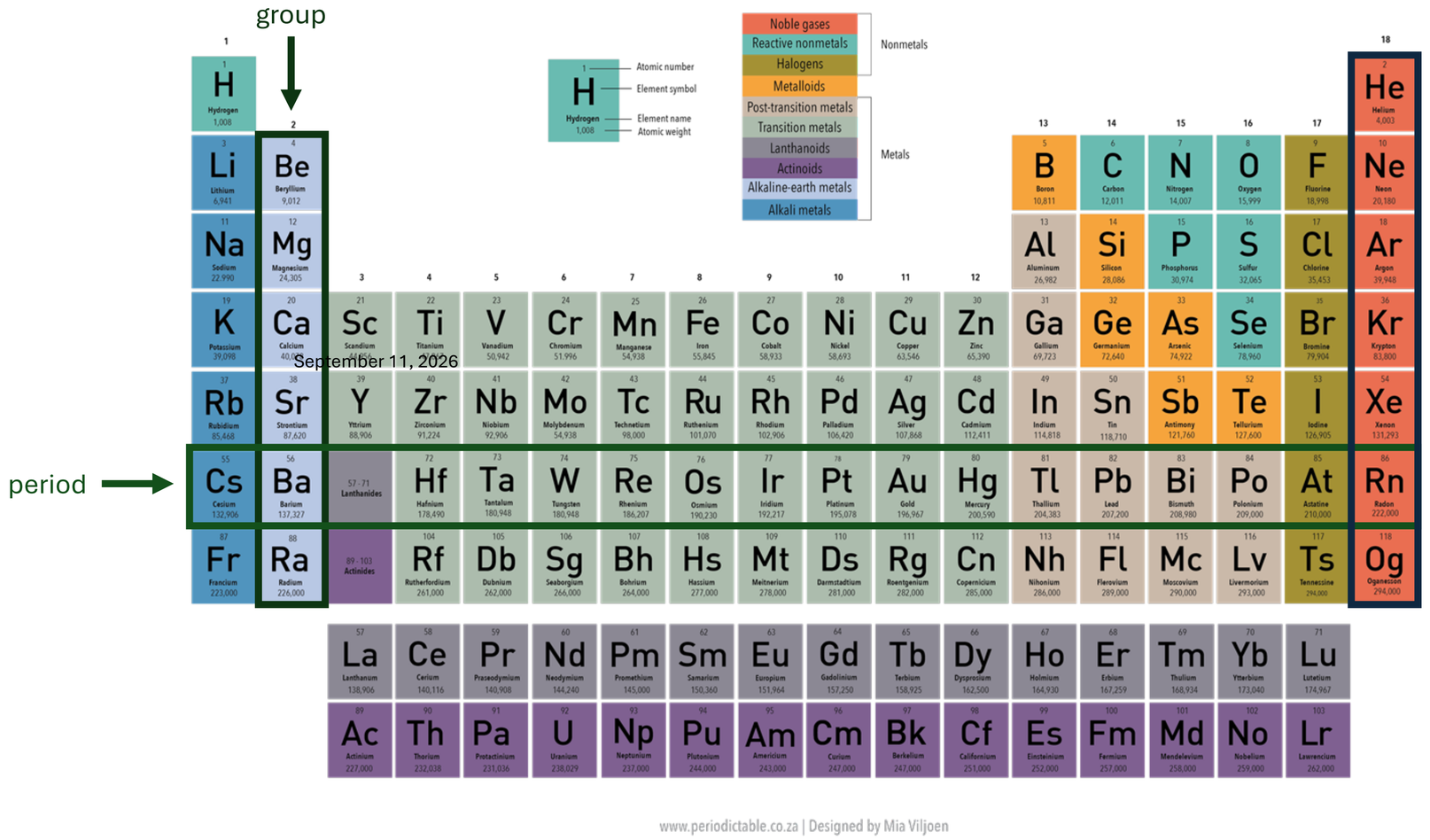


Solute Solvent Solution

Periodic table of the elements

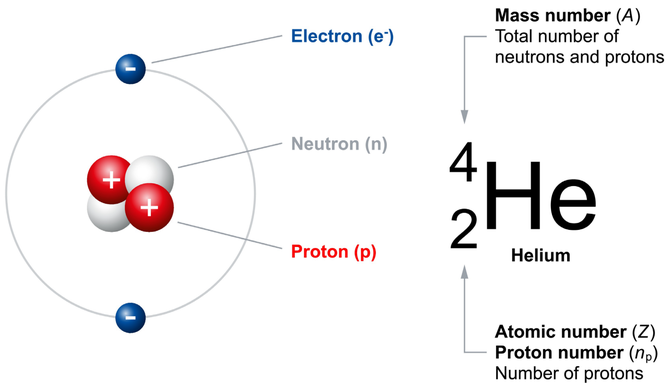






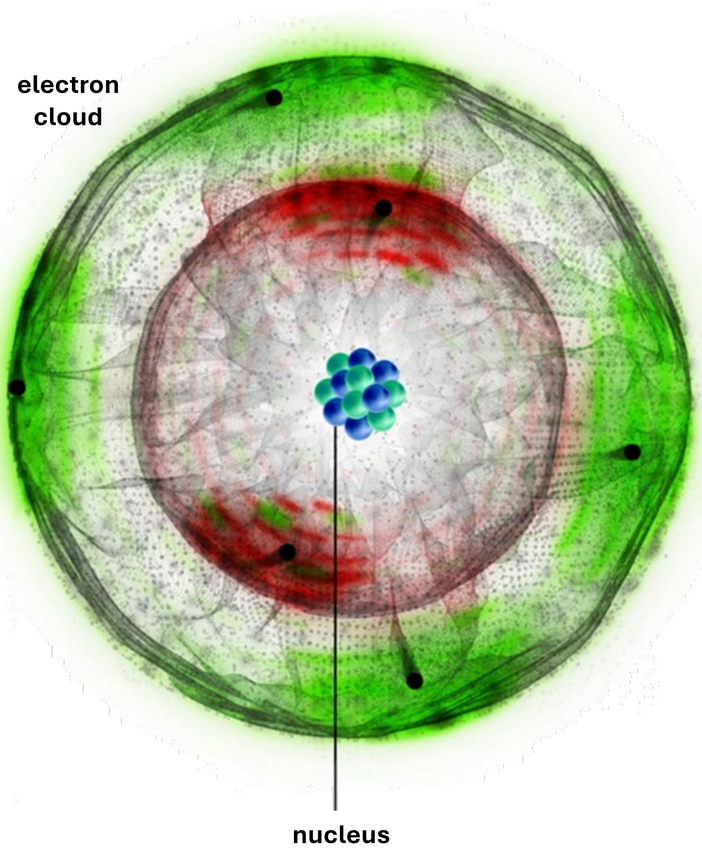
Atomic Mass

Property	Atomic Mass	Atomic Weight
Definition	Mass of a specific isotope	Average mass of all isotopes
Units	amu	No units (dimensionless) or amu
Specificity	Specific to one isotope	Weighted average for the element
Varies by isotope	Yes	Yes (but as a function of abundance)
Found in periodic table?	No (unless isotope table)	Yes



Mass number, A = no. protons (Z) + no. neutrons
Atomic mass = total mass of atom in amu (u)

Atoms: the building blocks of matter



Mass number (A)
Total number of neutrons and protons

$^{4}_{2}He$

Helium

Atomic number (Z)
Proton number (n_p)
Number of protons

+ **proton**
- positive charge (+1)
- 1.673×10^{-24} g = 1.0073 u
- atomic number, Z = no. protons

- **electron**
- negative charge (-1)
- 9.109×10^{-28} g = 0.0005486 u
- no. electrons = no. protons

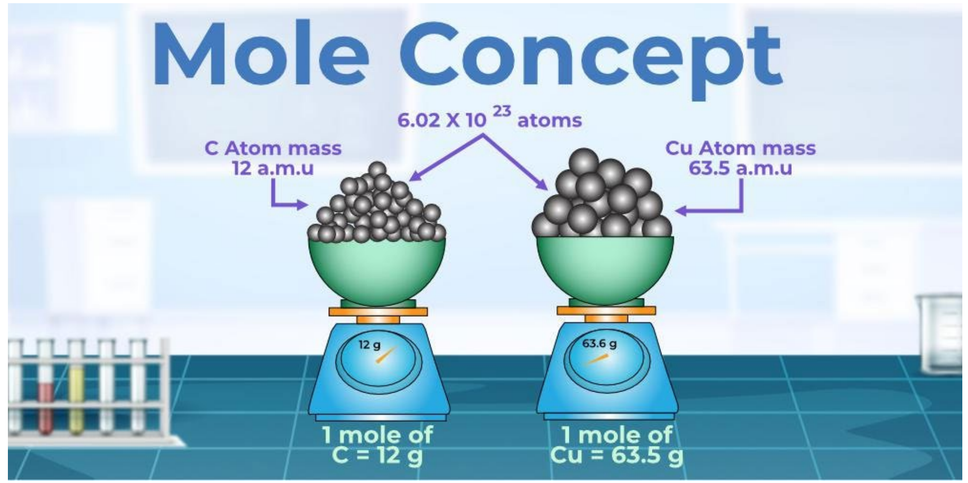
n **neutron**
- electrically neutral (0)
- 1.675×10^{-24} g = 1.0087 u
- A-Z = no. neutrons

Mass number, A = no. protons (Z) + no. neutrons
Atomic mass = total mass of atom in amu (u)

Amount of Substance n (Mole

Definition: How many particles are in one gram of a substance

6.022×10^{23} = Avogadro's Constant (N_a)



-One mole of an element contains 6.022×10^{23} atoms

-One mole of a compound contains 6.022×10^{23} molecules

Amount of Substance (n) = mol

- 1 mol = Relative atomic or molecular mass in grams

Formula: Molar mass (M) = mass (m) / amount of substance (mol)

$$M = \frac{m}{n}$$

Examples

Sodium:1 mol (n) weighs 23 g (m)→ Molar mass

(M) = m / n = 23 g/mol

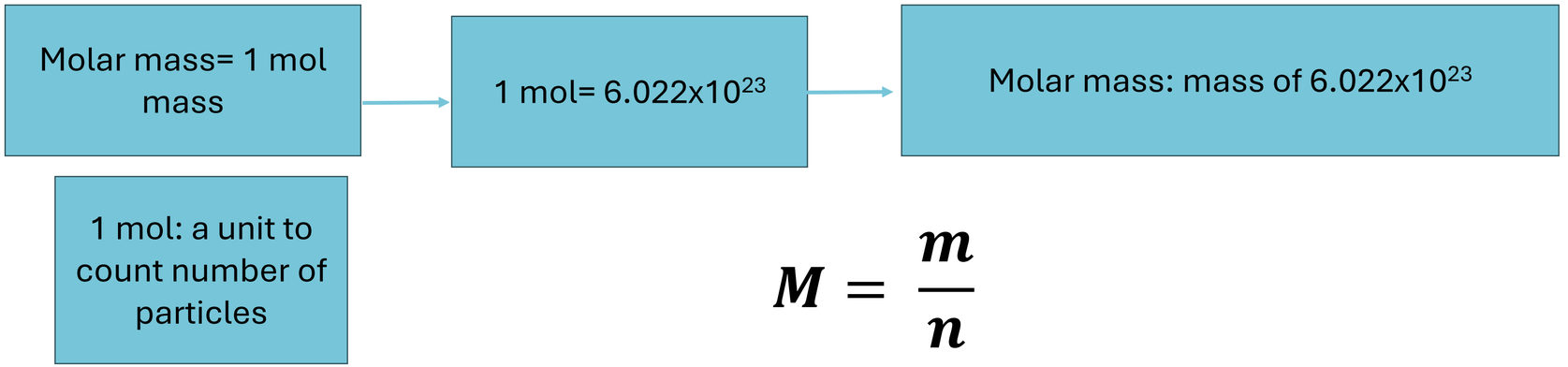
$$M = \frac{18\ g\ Na}{1\ mol} = 18\ g/mol$$

Water: H_2O: 1 mol weighs 18 g→ Molar mass = 18 g/mol

$$M = \frac{23\ g\ H_2O}{1\ mol} = 23 g/mol$$

Molar mass relation to avogadro constant

For compounds, the molecular mass (in amu) is numerically the same as the mass of one mole of the compound in grams.



$$M = \frac{m}{n}$$

Example: 1 mole of Aluminum: Al

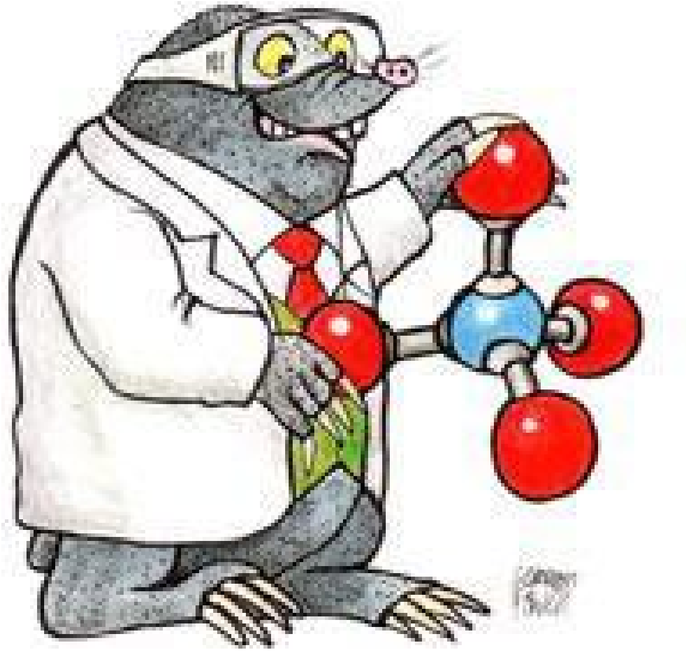
1 mol Alumium has a mass of 26 g
1 mol of Aluminum contains 6.022×10^{23} atoms
6.022×10^{23} Atom of Aluminium has mass of 27 g

Molecular mass of H_2O= (2 x atomic mass of H) + atomic mass of O
= 2 (1.008 amu) + 16.00 amu = 18.02

1 mole of H_2O= 6.022×10^{23} has a mass of 18.02 g.

The Mole

- 1 mol = 6.022×10^{23} particles (Avogadro constant).
- Based on carbon-12 definition.
- Molar mass = mass per mole.
- Example: C = 12 g/mol, H_2O = 18 g/mol



Molar mass vs Molecular mass

1. Write down the chemical formula of the molecular or ionic compound
2. Look up the atomic masses of the elements
3. Multiply the atomic masses with the corresponding **coefficient**
4. Add all the numbers

Covalent compound

Molecular mass H_2O = **2** (atomic mass H) + 1 (atomic mass O) = **2** (1.008 amu) + 15.999 amu = 18.015 amu

II

$M(H_2O)$ = 18.015 g/mol

Ionic compound

Formula mass $MgCl_2$ = atomic mass Mg + **2** (atomic mass Cl) = 24.305 amu + **2** (35.453 amu) = 95.211 u

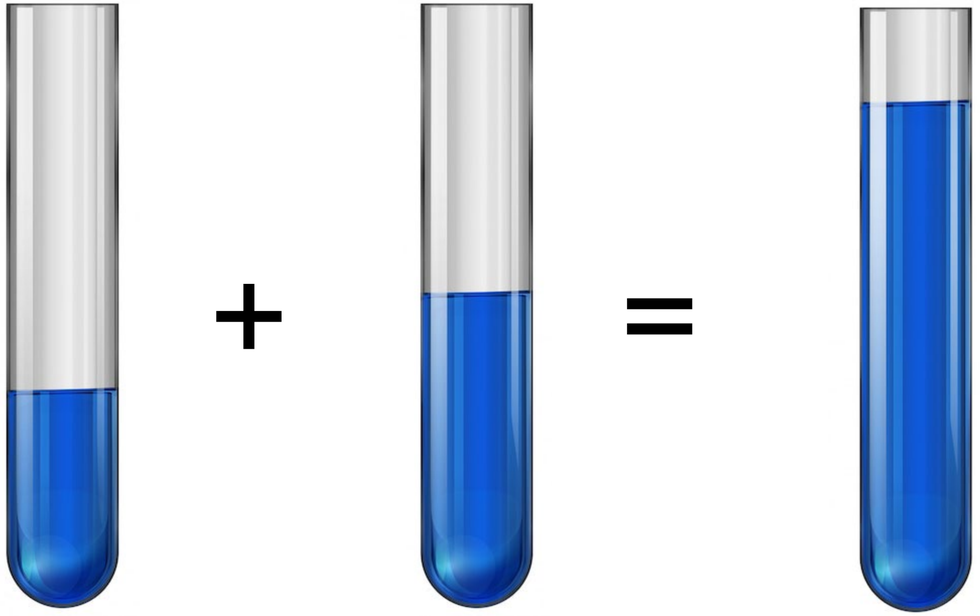
II

$M(MgCl_2)$= 95.211 g/mol

molar mass (M) = mass of 1 mole

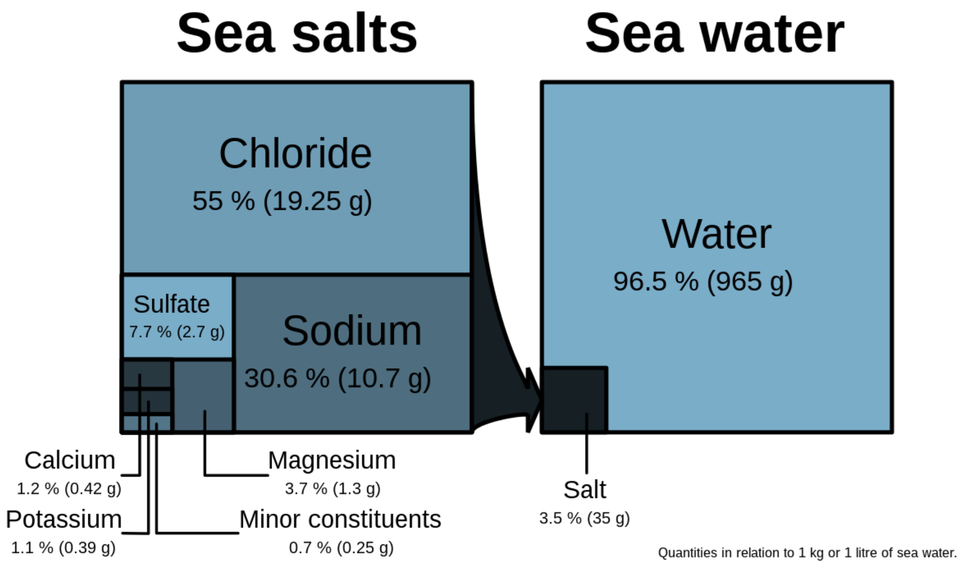
By convention, the **numerical value in amu per molecule** equals the **numerical value in g/mol per mole**.

Solutions



Solute + Solvent = Solution





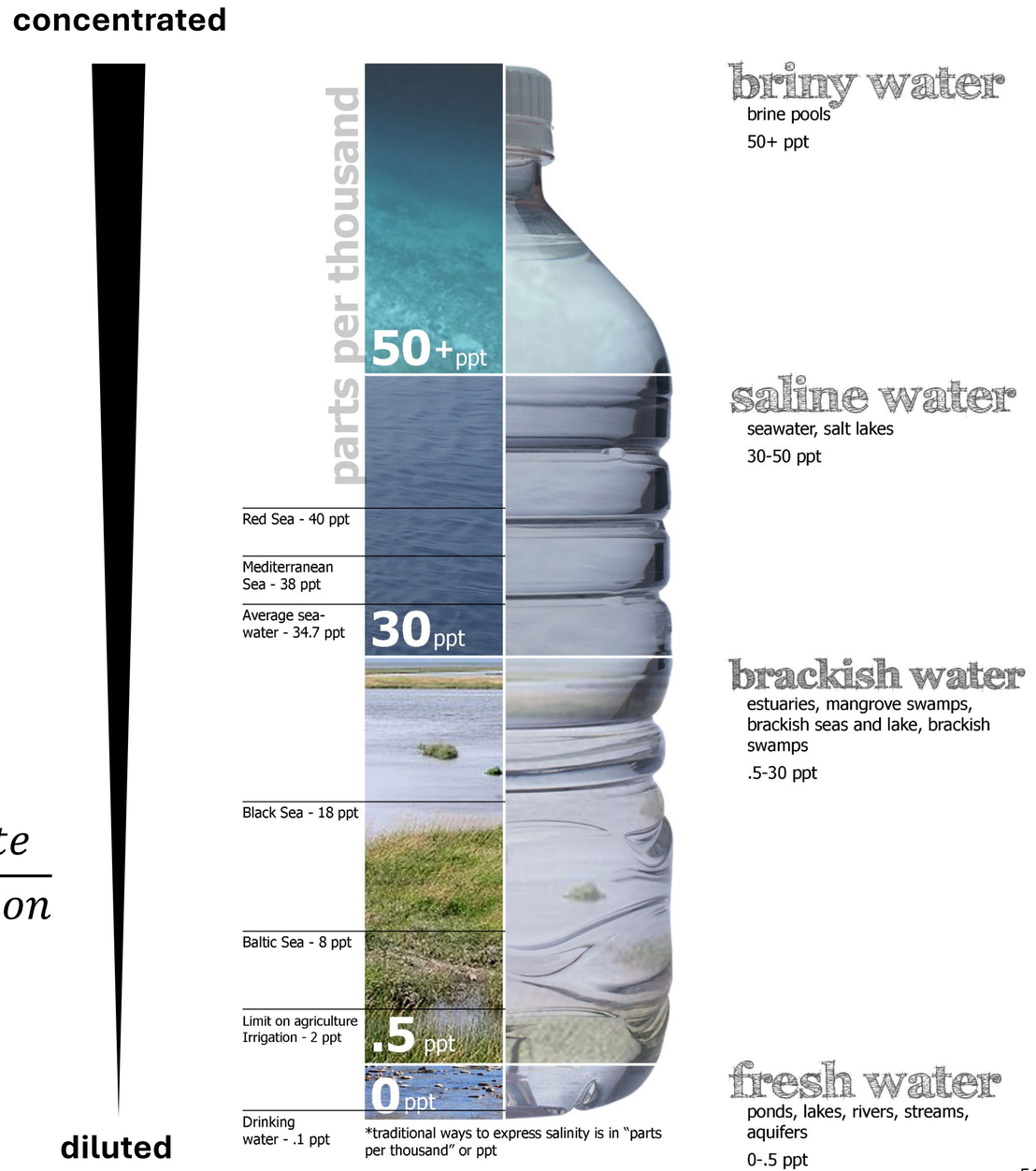
Solution concentration

$$Molarity\ (M) = \frac{n}{V} = \frac{moles\ solute}{L\ solution}$$

$$mass\ percentage\ (\%m\ or\ \%) = 100\ * \frac{g\ solute}{g\ solution}$$

$$volume\ percentage\ (\%V\ or\ \%Vol) = 100\ * \frac{L\ solute}{L\ solution}$$

$$ppm = \frac{mg\ solute}{kg\ solution} \quad ppb = \frac{\mu g\ solute}{kg\ solution} \quad ppt = \frac{ng\ solute}{kg\ solution}$$



Molarity

- Molarity (M): the amount of solute dissolved in a given amount of solution (not a given amount of solvent)

$$\textbf{Molarity (C)} = \frac{moles\ of\ Solute}{Liters\ of\ Solution} \qquad \textbf{C} = \frac{n}{V}$$

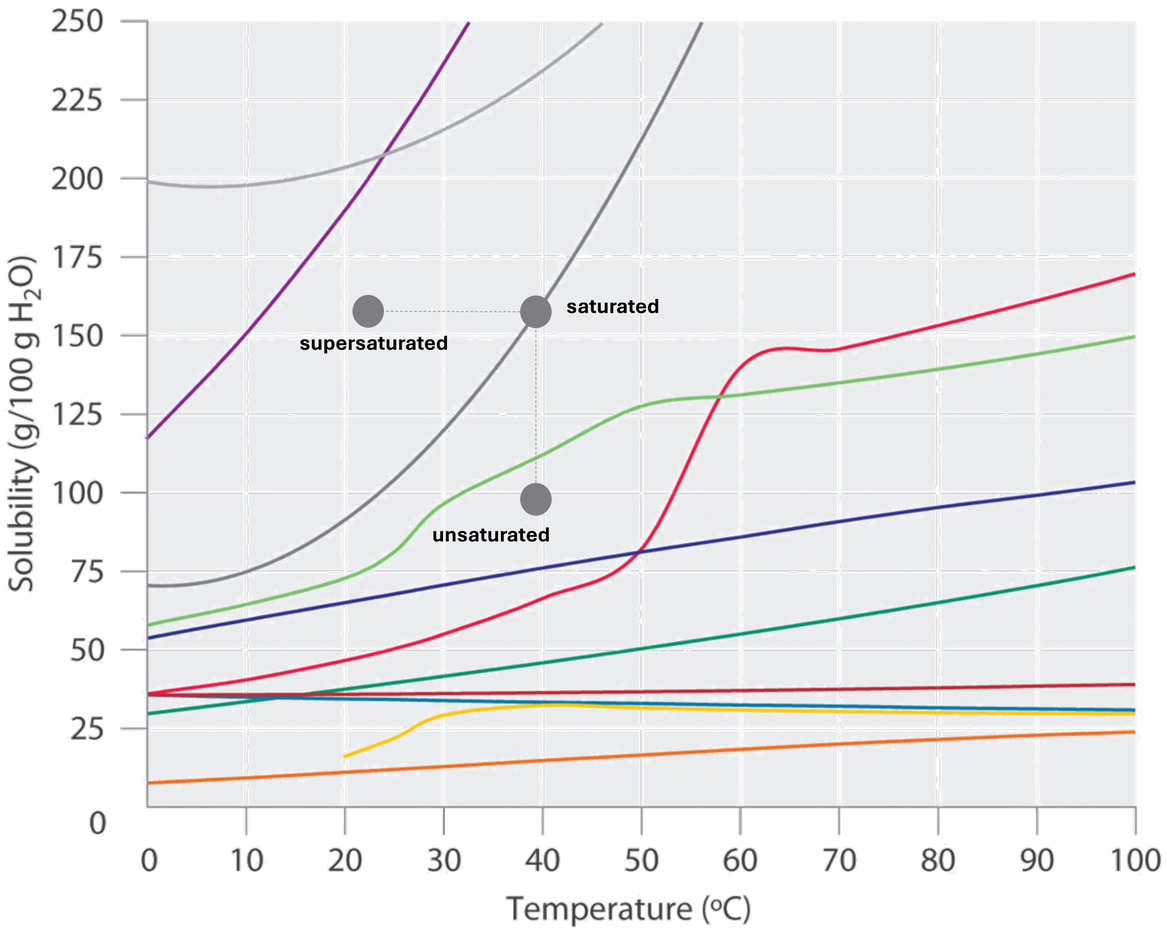
How to solve molarity problems?:

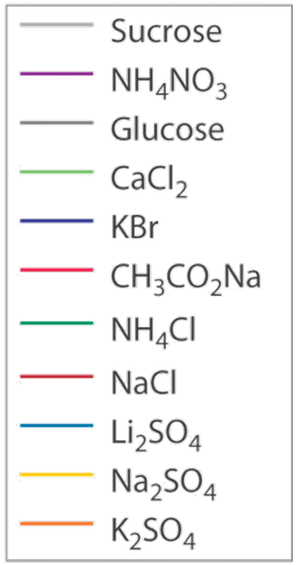
1. If you are given grams of solute, you **MUST** convert it into moles (divide by the molar mass). If given moles, go to step 2.

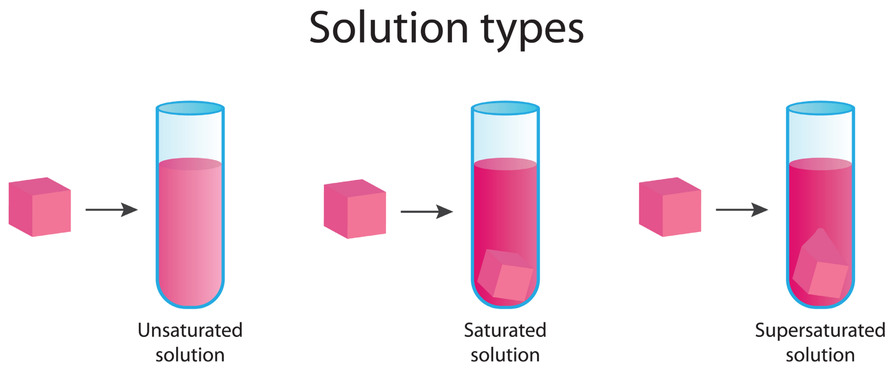
2. If you are given milliliters (mL) of the solute, you must convert into liters. (Divide by 1000). If given in liters go to step 3.

3. Use the formula to solve

Solubility







Solubility increases with increasing temperature: Kinetics

Solubility of gases decreases with increasing temperature: solvent molecules have more Kinetics

Dilutions

The **mass (m)** or **amount of substance (n)** remains constant, only the **amount of solvent increases** (i.e., **volume V changes**).

Formulas

Mass concentration:

C=m/V

Amount of substance concentration:

C=n/ V

Dilution Equation

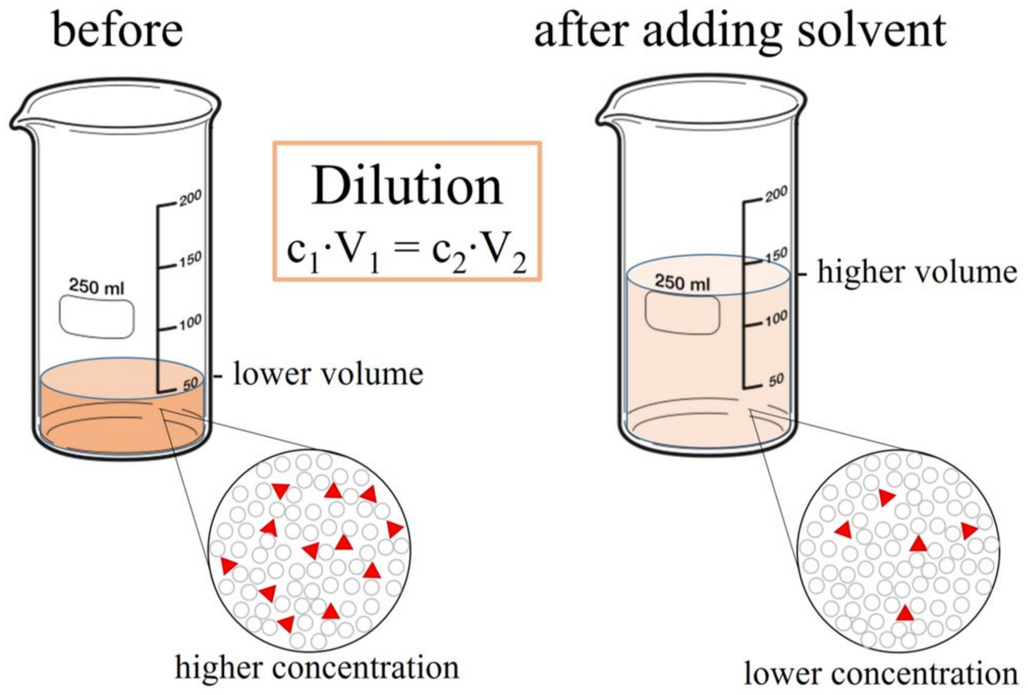
m1=m2 rn1=rn2 :

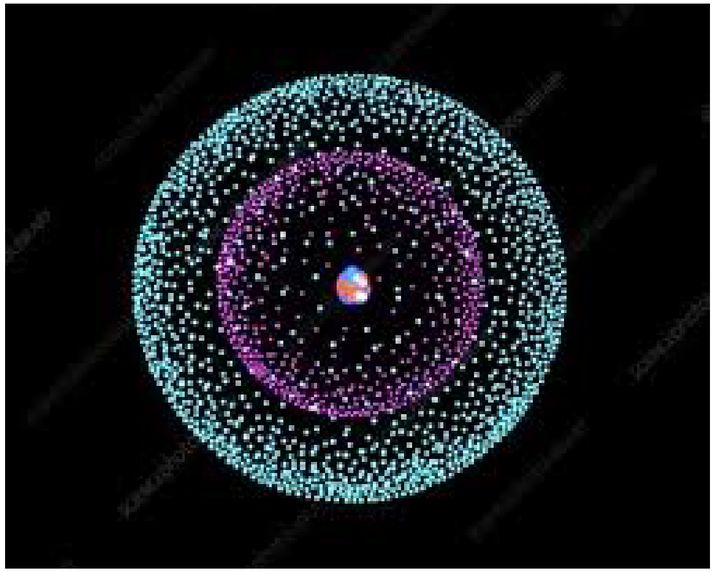
Formula: $c_1V_1 = c_2V_2$

Where:

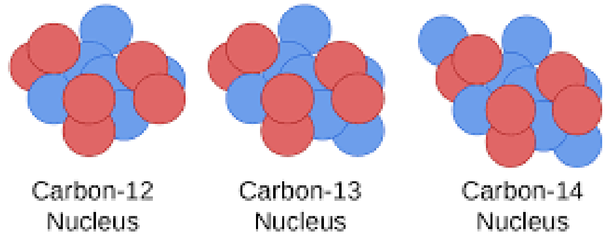
c_1 = initial concentration **V_1** = initial volume

c_2 = final concentration **V_2** = final volume









Outlook for Next Week:

- Structure of Atoms
- Atomic nucleus
- Atomic mass
- Isotopes
- Radioactivity